Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [13EC663/12EC663]

SIXTH SEMESTER B.TECH/B.E UNIVERSITY EXAMINATION, JUNE – 2016

DIGITAL SWITCHING SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	Explain the network services for Telecommunication.	04
	b	What is necessity of dividing and exchange into subdivision? Explain	04
	c	With neat sketch explain cross bar switching and its operation.	06
2.	a.	What is message switching and circuit switching? Explain.	08
	b.	Briefly explain the different network structures used in communication.	06
3.	a.	Define:	10
		i. Traffic iii. Lost call system v. Holding time	
		ii. Congestion iv. Erlang	
	b.	What is queuing system?	04
4.	a.	What is grading? Where and when is grade of service measured?	10
	b.	Write a note on:	04
		i. Progressive grading	
		ii. skipped grading system	
5.	a.	Discuss the need for frame alignment in time division switching networks	06
	b	Explain double ended unilateral and bilateral synchronization system.	08
6.	a.	With a neat diagram explain all 3 levels of software control architecture of a Switching Syste.	10
	b	Write a note on office data specifying the software and hardware parameters of digital switching system	O4
7.	a.	With table explain the switching system maintainability.	06
	b.	With a neat block diagram explain the interface of a typical switching system central office.	08
8.	a.	With a neat sketch explain the generic hardware architecture.	08
	b.	What are the common characteristics of digital switching systems.	06